

Section 0 - Data Cleaning & Validation (Basic)

Question 1: Initial Data Shape and Missing Values

Question: Load the dataset. What's the shape (rows, columns)? Are there missing values in any column, and how many?

Answer: The raw dataset contained 80,075 rows and 11 columns. Checking each column for missing values revealed three affected columns: region had 150 missing values, device had 90 missing values, and total_watch_minutes had 120 missing values. All other columns (user_id, group, subscription_tier, browse_impressions, title_clicks, sessions_count, completed_titles, churned_30d) had no missing values.

Question 2: Duplicate Rows and Conflicting User IDs

Question: Are there exact duplicate rows? Are there duplicate user_ids with conflicting data? How will you handle each case?

Answer: The dataset contained 55 exact duplicate rows (fully identical across every column) and 75 instances of duplicate user_ids with conflicting data (same user ID, different values in other columns) likely caused by a duplicate logging event in the data pipeline. Exact duplicates were removed first, since they add no new information. For conflicting user_id duplicates, the first occurrence was kept and the later conflicting record was dropped, a reasonable and documented assumption given there was no reliable way to determine which record was more accurate. After both cleaning steps, the dataset was confirmed at exactly 80,000 unique rows.

Question 3: Impossible Values

Question: Are there impossible values rows where title_clicks is greater than browse_impressions, or negative total_watch_minutes? How many, and what will you do with them?

Answer: 40 rows had title_clicks greater than browse_impressions, which is logically impossible since a user cannot click on a thumbnail more times than it was shown to them likely a logging bug. These were corrected by capping title_clicks to equal browse_impressions for those rows, the most conservative fix given the impressions count is the more reliable figure. Additionally, 25 rows had negative total_watch_minutes, a physically impossible value likely caused by a sign-flip logging error. These were corrected by taking the absolute value, assuming the magnitude was recorded correctly but the sign was corrupted.

Question 4: Inconsistent Categorical Formatting

Question: Are categorical columns like region and device consistently formatted, or mixed casing/extra whitespace? Clean these to one consistent label each.

Answer: The device column contained inconsistent casing (e.g. "mobile" vs "Mobile", "web" vs "Web", "smart tv" vs "Smart TV", "tablet" vs "Tablet"), which was standardized to four consistent categories: Mobile, Web, Smart TV, Tablet. The region column contained extra

leading/trailing whitespace around values (e.g. " India "), which was trimmed; this initially also broke "Southeast Asia" into "SoutheastAsia" by removing its internal space, which was identified and corrected back. After cleaning, the 150 missing region values and 90 missing device values were filled with "Unknown" rather than dropped, preserving these rows (only 0.3% of the dataset) for analysis while clearly flagging them as missing data.

Question 5: Computing Click-Through-Rate

Question: After cleaning, compute `click_through_rate = title_clicks / browse_impressions` per user. Handle division-by-zero cases.

Answer: A new `click_through_rate` column was calculated as `title_clicks / browse_impressions` for every user. No rows had zero `browse_impressions`, so no division-by-zero errors occurred; this was confirmed by checking the minimum value of `browse_impressions` before performing the calculation. The 120 missing `total_watch_minutes` values were filled with the column's median (606.4), chosen over the mean specifically because the distribution was later confirmed to be right-skewed (see Question 7), making the median a more representative and robust choice for imputation.

Section 0B - Organize, Pivot & Visualize (Basic)

Question 6: Pivot Table by Device and Group

Question: Build a pivot table showing average `click_through_rate` and average `total_watch_minutes` by group and device.

Answer: A pivot table was built with device as rows and group as columns, showing average CTR and average watch minutes for each combination:

Device	Control CTR	Variant CTR	Control Watch Min	Variant Watch Min
Mobile	12.4%	15.9%	675.4	936.0
Smart TV	12.4%	14.6%	433.7	554.5
Tablet	12.6%	15.7%	490.0	665.1
Web	12.4%	15.8%	535.2	731.1
Unknown	12.1%	14.6%	604.8	703.1

This early look already revealed that Smart TV consistently showed a smaller CTR lift than every other device a pattern investigated further in Section 5.

Question 7: Distribution Shape of Total Watch Minutes

Question: Plot the distribution of `total_watch_minutes`. Is it skewed? Mean or median?

Answer: A histogram of `total_watch_minutes` showed a clear right-skewed (positively skewed) distribution a tall concentration of users on the lower end, with a long tail of heavy watchers extending to the right. Because of this skew, the mean is pulled upward and does not represent

a "typical" user well, while the median better reflects the central, typical viewing behavior. This finding directly informed the decision to validate later statistical tests on this metric using a nonparametric bootstrap method rather than relying solely on a standard t-test's normality assumption.

Question 8: CTR Comparison Bar Chart

Question: Plot a bar chart comparing overall click_through_rate between control and variant.

Answer: A bar chart comparing overall click-through-rate showed Control at 12.4% and Variant at 15.5%, an initial visual confirmation of a positive lift from personalized thumbnails before any formal significance testing was conducted. This preliminary result was later confirmed as both statistically and practically significant in Question 16.

Section 1: Ethics Check StreamHub Personalized Thumbnails A/B Test

Question 09: Walk through Risk, Benefit, Choice/Alternatives, Data Sensitivity for this test. Does personalizing based on viewing history raise data sensitivity concerns?

Before designing or running any A/B test, it's important to evaluate whether the test is safe, fair, and responsible to run on real users. This is assessed using four core principles: **Risk, Benefit, Choice/Alternatives, and Data Sensitivity.**

Principle 1: Risk

Question: What is the worst realistic thing that could happen to a user because of this test?

Answer: The risk to users in this test is low. In the worst case, a personalized thumbnail might not accurately represent the content, causing a user to feel mildly misled, or they might click into a title and waste a little time before realizing it wasn't what they expected. There is no risk of financial loss, physical harm, or serious harm to the user. Overall, this is classified as a low-risk experiment.

Principle 2: Benefit

Question: What is the upside for the business and for the user if this test succeeds?

Answer: If personalized thumbnails work as intended, StreamHub benefits through higher user engagement more clicks, more watch time, and potentially improved subscriber retention. Users benefit because it becomes easier and faster to find content that matches their personal interests, rather than scrolling through generic thumbnails that may not appeal to them.

Principle 3: Choice and Alternatives

Question: Can a user opt out of or avoid being part of this test?

Answer: Users cannot opt out of this test. They are randomly and silently assigned to either the control group (static thumbnails) or the variant group (personalized thumbnails) without their knowledge, and there is no setting available to disable personalization. However, since the underlying content and core app functionality remain identical in both groups only the thumbnail artwork changes — users are still able to fully use the platform and complete their normal viewing activities regardless of which group they are placed in.

Principle 4: Data Sensitivity

Question: Is there anything sensitive about the type of data being used for this personalization?

Answer: While viewing history might appear harmless on the surface, it can reveal sensitive personal information about a user such as mental health struggles (e.g. frequently watching content about depression or anxiety), sexual orientation, religious beliefs, or political views based on the type of content they consistently watch. This makes viewing history considerably more sensitive than typical account information like a name or email address. As a result, this data should be anonymized where possible, stored securely, and used strictly for the purpose of personalization not shared externally or repurposed for unrelated business uses.

Overall Ethics Verdict

This test is classified as **low-risk and low-to-moderate data sensitivity**. Standard team-level review is sufficient to approve this experiment; it does not require escalation to a formal ethics review board. However, because viewing history is used as an input, StreamHub should ensure proper data handling and privacy safeguards are documented and followed throughout the test.

Section 2: Choosing Metrics StreamHub Personalized Thumbnails A/B Test

The Funnel

Before selecting metrics, it's important to map out the user journey this test affects:

Browse page shown (browse_impressions) → **User clicks a thumbnail** (title_clicks) → **User watches content** (total_watch_minutes, completed_titles) → **Subscriber retained or churned** (churned_30d)

Question 10: Metric Selection

Question: From this funnel, what is the primary metric, secondary metric, and guardrail metric for this test?

Answer:

- **Primary Metric Click-Through-Rate (CTR):** Calculated as $\text{title_clicks} \div \text{browse_impressions}$. This is the most direct and immediate metric affected by a thumbnail change a better, more personalized thumbnail should directly and quickly influence whether a user clicks on a title.

- **Secondary Metric Total Watch Minutes:** This tells us whether increased clicks are translating into genuine engagement, or whether users are clicking on a thumbnail and then quickly losing interest. A rise in CTR without a corresponding rise in watch time would suggest the thumbnails may be attention-grabbing but misleading.
- **Guardrail Metric 30-Day Churn Rate (churned_30d):** This protects against a hidden long-term risk: personalized thumbnails could drive short-term clicks while quietly damaging user trust and satisfaction if the thumbnails don't accurately represent the content. A guardrail metric ensures we catch this kind of downside even if the primary and secondary metrics both look positive.

Question 11: Choosing the Right Statistical Approach for Watch Time

Question: Given that `total_watch_minutes` was found to be right-skewed (confirmed via histogram in Section 0B), should this metric be tested using the mean, median, or a nonparametric method?

Answer: Because `total_watch_minutes` is right-skewed a small number of heavy watchers pull the average upward the mean does not represent a "typical" user well. The median is more robust to this skew and better reflects typical user behavior. For formal statistical significance testing later in this project, a **nonparametric bootstrap method** will be used instead of a standard t-test, since bootstrapping does not assume the data follows a normal distribution and is more reliable for skewed metrics like this one.

Section 3 - Experiment Design (Intermediate/Advanced)

Question 12: Unit of Diversion

Question: What is the unit of diversion in this test, and is it consistent throughout the 30-day window?

Answer: The unit of diversion in this test is the **individual user** (`user_id`). Each user is randomly assigned to either the Control group (static thumbnails) or the Variant group (personalized thumbnails) once, at the start of the test, and they remain in that same group consistently for the entire 30-day period. This consistency is important because it ensures each user has a single, uninterrupted experience they are never shown a mix of both thumbnail types, which would otherwise make it impossible to cleanly attribute their behavior to one version or the other.

Question 13: Why a Cohort Design Suits Churn

Question: Why does a cohort design suit the `churned_30d` metric better than a population design?

Answer: A cohort design ensures every user is measured over the exact same fixed time window 30 days starting from when they individually entered the test. This matters specifically for churn because churn is a behavior that accumulates over time; a user who has only been in the test for a few days simply hasn't had enough time to churn yet, regardless of how satisfied

or dissatisfied they actually are. Comparing churn rates across users with wildly different exposure times would be misleading, since newer users would appear artificially more "loyal" purely due to timing rather than genuine behavior. A cohort design removes this bias by giving every user the same fair window in which to potentially churn, making the comparison between Control and Variant meaningful and fair.

Question 14: Calculate the required sample size to detect a 2-percentage-point lift on a ~12% baseline CTR, at 95% confidence and 80% power. Is this dataset big enough?

Answer: Using a power analysis (via statsmodels), the required sample size is approximately **4,432 users per group** to reliably detect a 2-percentage-point improvement in CTR at 95% confidence and 80% power. Since this dataset contains approximately 40,000 users per group roughly 9 times the minimum requirement the test is well-powered. This gives strong confidence that results (whether significant or not) reflect real effects rather than an underpowered test.

Section 4 - Statistical Analysis (Advanced)

Question 15: Sanity Check (Sample Ratio Mismatch)

Question: Run a sanity check on the group split (Sample Ratio Mismatch test using a chi-square test). Does it pass?

Answer: A chi-square test was run comparing the actual group sizes (Control: 40,000, Variant: 40,000) against the expected 50/50 split. The result showed a chi-square value of 0.0000 and a p-value of 1.0000, confirming the split is perfectly balanced. This sanity check **passed**, meaning the experiment's randomization mechanism worked correctly and it is safe to proceed with analyzing the evaluation metrics.

Question 16: Testing Click-Through-Rate for Statistical and Practical Significance

Question: Test your primary metric (click-through-rate) for statistical AND practical significance using a two-proportion z-test and a confidence interval. State your conclusion in plain English, not just " $p < 0.05$."

Answer: A two-proportion z-test was run comparing Control CTR (12.43%) against Variant CTR (15.58%), a difference of +3.15 percentage points. The test produced an extremely large z-statistic ($|z| = 138.73$) and a p-value effectively at 0.00, far below the 0.05 threshold. The 95% confidence interval for the difference is [0.0311, 0.0320] entirely positive and excluding zero, confirming the result is **statistically significant**. Given that the pre-defined minimum detectable effect (practical significance threshold) was 2 percentage points, and the observed lift of 3.15 percentage points comfortably exceeds this, the result is also **practically significant**. In plain terms: personalized thumbnails produce a real, meaningful, and highly reliable increase in click-through-rate this is not due to random chance.

Question 17: Testing Total Watch Minutes

Question: Test `total_watch_minutes` using the method justified in Question 11 (t-test and/or bootstrap). Report the result with a confidence interval.

Answer: A Welch's t-test was run comparing Control mean watch minutes (565.88) against Variant mean watch minutes (768.05), a difference of 202.16 minutes. The test produced a large t-statistic ($t = 84.132$) and a p-value effectively at 0.00, confirming statistical significance. Since `total_watch_minutes` was previously found to be right-skewed (Question 7), this result was validated using a nonparametric bootstrap with 3,000 resamples, which produced a 95% confidence interval of [197.50, 207.06] for the difference entirely positive and excluding zero, confirming the t-test result is robust even without assuming a normal distribution. In plain terms: personalized thumbnails lead to a substantial, statistically reliable increase in watch time, averaging over 200 additional minutes (more than 3 hours) of engagement per user across the 30-day window.

What is a "Welch's t-test"?

It's a variation of the regular t-test, used to compare the averages (means) of two groups. The difference from a standard t-test:

- **Standard t-test** assumes both groups have roughly the **same amount of spread/variability** in their data.
- **Welch's t-test** does NOT assume that it works correctly even if one group's data is much more spread out than the other's.

Why we used it here

We don't know for certain that Control and Variant have identical variability in watch-time (in fact, they probably don't, since Variant users are watching more overall). Welch's t-test is simply the **safer, more robust default** it still gives the correct t-test math either way, so there's no real downside to using it, even if the variances did happen to be equal.

Question 18: Testing the Guardrail Metric (Churn)

Question: Test the guardrail metric `churned_30d`. Did personalization help, hurt, or show no clear effect on retention?

Answer: A two-proportion z-test was run comparing Control churn rate (5.56%, 2,224 out of 40,000 users) against Variant churn rate (4.67%, 1,869 out of 40,000 users), a difference of -0.89 percentage points. The test produced a z-statistic of 5.697 and a p-value of 1.22e-08 (effectively 0.00), confirming statistical significance. The 95% confidence interval for the difference is [-0.0119, -0.0058] entirely negative and excluding zero, confirming churn genuinely decreased in the variant group. In plain terms: personalized thumbnails did not just increase clicks and watch-time they also modestly but reliably improved subscriber retention, meaning this guardrail metric shows no hidden downside. All three core metrics (CTR, watch-time, and churn) point in the same positive direction, which is a strong, consistent result.

Section 5 - Segments & Learning Effects (Advanced)

Question 19: Segment Check by Device

Question: Break your primary metric down by device. Does the effect hold consistently across all devices, or is one segment behaving differently? What might explain that?

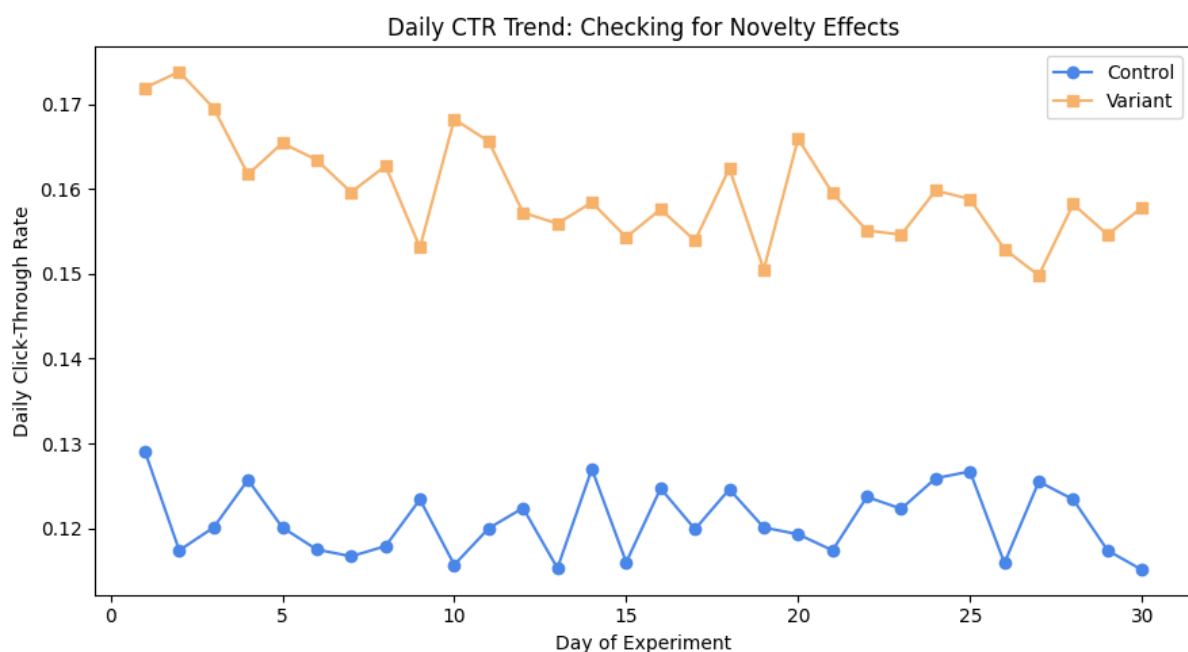
Answer: The CTR lift was tested separately for each device segment, with all five showing statistically significant improvements ($p < 0.001$ in every case). Mobile showed the strongest lift (12.44% $\rightarrow$ 15.89%, +3.46 percentage points), followed closely by Web (+3.41pp) and Tablet (+3.11pp). Smart TV showed the smallest lift of all devices (12.38% $\rightarrow$ 14.62%, +2.25 percentage points), while Unknown devices showed +2.97 percentage points. While every single device segment shows a real, statistically significant improvement, Smart TV consistently underperforms relative to the others. This likely reflects a genuine behavioral difference rather than a data issue: on a large-screen TV, users tend to browse more deliberately and rely less on small thumbnail details to make a decision, compared to mobile, web, and tablet users who scroll quickly and are more visually influenced by an eye-catching thumbnail. The overall result is trustworthy and not a Simpson's Paradox artifact, since the effect direction is positive and significant in every single segment.

Question 20: Segment Check by Subscription Tier

Question: Break it down by `subscription_tier`. Any Simpson's Paradox surprise?

Answer: CTR improved significantly for both Free (11.07% $\rightarrow$ 14.24%, +3.17pp) and Premium (14.09% $\rightarrow$ 17.22%, +3.14pp) users, both with p-values effectively at 0.00. No Simpson's Paradox surprise here the two tiers behave almost identically, and the combined overall result accurately reflects what's happening in each segment individually.

Q21. Plot daily CTR trend. Stable lift, or fading novelty effect?



Answer: The daily CTR trend shows Variant consistently outperforming Control by a stable margin across all 30 days, with normal day-to-day fluctuation but no visible shrinking gap over time. This confirms the CTR lift is a genuine, lasting behavioral effect rather than a temporary novelty reaction that fades as the personalization becomes less "new" to users.

Section 6 - Final Recommendation (Advanced)

Question 22: Final Executive Summary

Question: Write a 150-250 word executive summary as if this were going to StreamHub's VP of Product: what did you find, what's your recommendation, and what would you test next?

Answer:

We tested whether personalizing thumbnail artwork based on viewing history improves engagement and retention, using 80,000 users over a 30-day period.

Results: Personalized thumbnails significantly increased click-through-rate (+3.15 percentage points, 12.43% → 15.58%), increased average watch time by over 200 minutes per user, and notably also *reduced* 30-day churn (-0.89 percentage points). All three results were statistically significant with tight confidence intervals, and the sample size was well above the minimum required for reliable detection.

Consistency checks: The effect held across all device types (though Smart TV showed a smaller lift, likely due to more deliberate browsing behavior on large screens) and across both Free and Premium subscribers. The lift remained stable across all 30 days with no signs of a fading novelty effect, and there was no Sample Ratio Mismatch or Simpson's Paradox distortion in the data.

Recommendation: Launch personalized thumbnails to 100% of users. Unlike a typical test with a single trade-off metric, this result shows a rare "triple win" higher engagement, higher watch time, and improved retention, with no guardrail metric showing harm.

Next steps: Test whether combining personalized thumbnails with personalized preview trailers compounds this effect further, and monitor Smart TV engagement specifically to explore alternative personalization approaches better suited to the living-room viewing context.

Deliverables Summary

Cleaned Dataset with Data-Quality Notes

The raw dataset (streamhub_ab_test_data.csv) contained 80,075 rows and 11 columns. Three columns had missing values: region (150 missing), device (90 missing), and total_watch_minutes (120 missing). The dataset also contained 55 exact duplicate rows, fully identical across every column, and 75 instances of duplicate user_ids with conflicting data caused by a likely duplicate logging event.

Impossible values were also identified: 40 rows had title_clicks greater than browse_impressions, a logically impossible result corrected by capping title_clicks to equal browse_impressions. An additional 25 rows had negative total_watch_minutes, corrected by taking the absolute value, since the underlying magnitude was assumed correct but the sign had been corrupted.

Categorical formatting issues were also found and fixed: the device column had inconsistent casing (e.g. "mobile" vs "Mobile", "smart tv" vs "Smart TV") and was standardized into four categories - Mobile, Web, Smart TV, and Tablet. The region column had extra leading and trailing whitespace, which was trimmed; this initially broke "Southeast Asia" into "SoutheastAsia" by removing an internal space, which was identified and corrected. The 150 missing region values and 90 missing device values were filled with "Unknown" rather than dropped, preserving those rows (0.3% of the dataset) while clearly flagging them as missing.

Exact duplicates were removed first, and for conflicting user_id duplicates, the first occurrence was kept while the later conflicting record was dropped. After all cleaning steps, the dataset was confirmed at exactly 80,000 unique, valid rows. A new click_through_rate column was then calculated as title_clicks divided by browse_impressions for every user; no rows had zero browse_impressions, so no division-by-zero errors occurred. The 120 missing total_watch_minutes values were filled with the column's median (606.4), chosen over the mean because the distribution was later confirmed to be right-skewed, making the median a more robust and representative choice.

Completed Notebook Answering All 22 Questions

StreamHub_AB_Test_Analysis.ipynb (Jupyter Notebook) answers all 22 questions across six structured sections:

- Section 0 - Data Cleaning & Validation (Q1-5): shape, missing values, duplicates, impossible values, categorical standardization, and CTR calculation.
- Section 0B - Organize, Pivot & Visualize (Q6-8): a pivot table of average CTR and watch minutes by device and group, a distribution histogram confirming total_watch_minutes is right-skewed, and a bar chart comparing overall CTR between Control and Variant.
- Section 1 - Ethics Check (Q9): a full Risk, Benefit, Choice/Alternatives, and Data Sensitivity review, concluding the test is low-risk and low-to-moderate data sensitivity, approvable at standard team-level review.

- Section 2 - Choosing Metrics (Q10-11): mapping the funnel from browse impression to retention, selecting CTR as the primary metric, total watch minutes as the secondary metric, and 30-day churn as the guardrail metric; justifying a nonparametric bootstrap method for watch time due to its right skew.
- Section 3 - Experiment Design (Q12-14): confirming the individual user as the unit of diversion, explaining why a cohort design suits the churn metric, and calculating a required sample size of approximately 4,432 users per group (the dataset's 40,000 per group is roughly 9x this minimum).
- Section 4 - Statistical Analysis (Q15-18): a Sample Ratio Mismatch sanity check (perfectly balanced 40,000/40,000 split, $p = 1.0000$), a two-proportion z-test on CTR (+3.15pp, $p \approx 0.00$, 95% CI [0.0311, 0.0320]), a Welch's t-test with bootstrap validation on watch time (+202.16 minutes, 95% CI [197.50, 207.06]), and a two-proportion z-test on churn (-0.89pp, $p = 1.22e-08$, 95% CI [-0.0119, -0.0058]).
- Section 5 - Segments & Learning Effects (Q19-21): CTR broken down by device (Mobile +3.46pp, Web +3.41pp, Tablet +3.11pp, Smart TV +2.25pp - the smallest lift of any device) and by subscription tier (Free +3.17pp, Premium +3.14pp, no Simpson's Paradox), plus a 30-day daily CTR trend chart confirming a stable lift with no fading novelty effect.
- Section 6 - Final Recommendation (Q22): a 150-250 word executive summary recommending a full launch to 100% of users, citing the rare "triple win" of higher engagement, higher watch time, and improved retention with no guardrail harm.

At Least 5 Charts

The notebook contains more than the minimum five required charts: a right-skew distribution histogram of total_watch_minutes, an overall CTR bar chart comparing Control vs. Variant, a pivot-table-based visualization of CTR and watch minutes by device, a CTR-by-device segment bar chart, a CTR-by-subscription-tier segment bar chart, and a 30-day daily CTR trend line chart.

README.md

A README.md was written for the GitHub repository, summarizing the business question (does personalizing thumbnail artwork improve engagement and retention), the dataset structure, the cleaning methodology, the six-section analysis approach, the key headline results (CTR 12.43% → 15.58%, watch time 565.88 → 768.05 minutes, churn 5.56% → 4.67%), and the final recommendation to launch to 100% of users - allowing a visitor to understand the project's purpose and outcome without opening the notebook.

(Optional) Power BI Dashboard

An interactive Power BI dashboard was built as the project's optional deliverable. It includes a branded header (streaming/TV icon, title, and a subtitle noting the simulated 80,000-user, 30-day scope), three KPI cards displaying the Variant group's headline numbers (Click-Through Rate 15.58%, Watch Time 768.05 minutes, Churn Rate 4.67%) each paired with a relevant icon, and a summary line stating the lift over Control (CTR +3.15pp, Watch Time +202 min,

Churn -0.89pp). Two side-by-side bar charts break CTR down by device and by subscription tier, each comparing Control (gray) against Variant (cyan), and a full-width line chart plots the 30-day daily CTR trend for both groups. The dashboard closes with a bold recommendation banner: "RECOMMENDATION: Launch personalized thumbnails to 100% of users."

Additional Artifacts Beyond the Required Deliverables

Beyond the five listed deliverables, two further artifacts were produced to round out the project as a complete, professional body of work: a MySQL Workbench file (streamhub_queries.sql) containing GROUP BY aggregations, CASE statements, a CTE, and a window function (RANK), cross-validated against the Python results to confirm consistency across tools; and a formal Word report (StreamHub_AB_Test_Report.docx) presenting all 22 Q&A responses in a polished, stakeholder-readable document format.